

GLP–Graph Scaling Behaviour

Cognitive Manifold Stability Across Generations

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Abstract

The GLP–Graph (GLPG) architecture is a structurally deterministic yet semantically emergent language substrate whose behaviour is governed by a stable attractor manifold. Whereas previous work characterised GLPG qualitatively, the present paper introduces the first numerical evaluation pipeline for GLPG: all generated stories across all scaling experiments are assessed by an external LLM using a calibrated scoring regime. This enables objective, reproducible comparison across model sizes, generative modes, training depths, staged lineages, and transfer-learning regimes.

Using this metric, we investigate GLPG under upscaling, downscaling, staged training, and cross-dataset inheritance. The results show that GLPG supports function-preserving scaling: the learned lemma embeddings and paging geometry act as cognitive DNA that remains stable across architectural transformations. Upscaling expands the semantic manifold without erasing existing attractors; downscaling compresses the manifold while preserving canonical structure; staged training produces hereditary attractor clusters that propagate across generations; and transfer learning reorganises the manifold in a deterministic, canon-dominant manner. Large models trained from scratch perform significantly worse than lineage-grown models, demonstrating that capacity alone is insufficient without prior consolidation.

Across all experiments, GLPG exhibits unusually strong structural and semantic stability. Its manifold deforms smoothly under hyperparameter variation, scaling, compression, and dataset transfer, behaving not as a statistical correlation space but as a low-dimensional cognitive dynamical system. The numerical evaluation introduced in this paper establishes GLPG as a measurable, interpretable, and evolution-capable cognitive substrate within the Triadic Cosmos ecosystem.

Significance Statement

The GLP–Graph (GLPG) architecture was originally introduced as a minimal and interpretable cognitive substrate capable of combining deterministic graph routing with emergent semantic manifolds. The present paper extends this foundation by demonstrating that GLPG is not merely a static architecture, but a scalable and evolution-capable cognitive system whose attractor geometry remains stable across architectural transformations, dataset transfers, and staged generational growth.

This work introduces the first numerical evaluation pipeline for GLPG: all generated stories across all scaling experiments are assessed by an external LLM using a calibrated scoring regime. This provides an objective and reproducible metric for comparing cognitive behaviour across model sizes, generative modes, training depths, staged lineages, and transfer-learning directions. The resulting measurements reveal that GLPG supports function-preserving scaling: the learned lemma embeddings and paging geometry act as cognitive DNA that survives both upscaling and downscaling, enabling models to grow or shrink without cognitive reset.

The experiments show that GLPG exhibits unusually strong structural and semantic stability. Upscaling inflates the semantic manifold while preserving canonical structure; downscaling compresses

the manifold without erasing core attractors; staged training produces hereditary semantic clusters that propagate across generations; and transfer learning reorganises the manifold in a deterministic, canon-dominant manner. Large models trained from scratch perform significantly worse than lineage-grown models, demonstrating that capacity alone is insufficient without prior consolidation at smaller scales.

These findings position GLPG as a minimal yet robust cognitive substrate. Its bidirectional scalability, hereditary attractor dynamics, stability under variation, and staged growth behaviour provide a concrete computational analogue for cognitive development, inheritance, and world-formation within the Triadic Cosmos ecosystem. GLPG therefore serves not only as a language generator, but as a structurally grounded model of scalable artificial cognition.

1 Introduction

The GLP-Graph (GLPG) architecture was introduced as a hybrid generative substrate that unifies deterministic graph routing with emergent semantic behaviour inside a single monolithic MLP. Whereas conventional language models rely on large-scale attention mechanisms and implicit statistical correlations, GLPG provides an explicit and inspectable governance layer: grammar tokens, lemma transitions, paging geometry, and structural cadence are defined by a canonical graph, while semantic content is injected through learned lemma embeddings and a compact feedforward network. This combination yields a system that is both structurally deterministic and semantically expressive, forming a minimal cognitive substrate within the Triadic Cosmos ecosystem.

Previous work established two foundational properties of GLPG. The architecture paper demonstrated that GLPG supports dual generative modes—paged graph traversal and pure MLP sampling—revealing that the monolithic network internalises manifold structure that mirrors the deterministic graph. The hyperparameter study showed that GLPG exhibits structured cognitive dynamics across scale: proto-semantic attractors in underscaled models, emergent semantic manifolds in mid-scale configurations, and canon compression in larger models. These behaviours suggest that GLPG operates not merely as a language generator, but as a low-dimensional dynamical system whose attractor geometry evolves predictably with capacity.

The present paper extends this line of inquiry by investigating GLPG under architectural scaling, staged generational growth, and cross-dataset transfer. We examine how the attractor manifold behaves when the hidden-layer capacity of the monolithic MLP is increased, decreased, or grown through lineage. Crucially, we introduce the first numerical evaluation pipeline for GLPG: all generated stories across all scaling experiments are assessed by an external LLM using a calibrated scoring regime. This provides an objective and reproducible metric for comparing cognitive behaviour across model sizes, generative modes, training depths, staged lineages, and transfer-learning directions.

Using this metric, we show that GLPG supports function-preserving scaling. The learned lemma embeddings and paging geometry act as cognitive DNA that remains stable across architectural transformations. Upscaling expands the semantic manifold without erasing existing attractors; downscaling compresses the manifold while preserving canonical structure; staged training produces hereditary attractor clusters that propagate across generations; and transfer learning reorganises the manifold in a deterministic, canon-dominant manner. Large models trained from scratch perform significantly worse than lineage-grown models, demonstrating that capacity alone is insufficient without prior consolidation.

Across all experiments, GLPG exhibits unusually strong structural and semantic stability. Its manifold deforms smoothly under hyperparameter variation, scaling, compression, and dataset transfer, behaving not as a statistical correlation space but as a structured cognitive dynamical system. This stability enables predictable cognitive development across architectural stages and provides a foundation for lineage-based world formation.

Overall, this paper positions GLPG as a scalable and evolution-capable cognitive substrate. Its bidirectional stability, hereditary attractor dynamics, transfer-learning behaviour, and staged growth properties provide a computational analogue for the AGI lifecycle concept: cognitive systems that develop, compress, inherit, and expand their internal world models across generations.

2 Related Work

Research on scaling behaviour in neural architectures has largely focused on transformers and their predecessors—MLPs, RNNs, LSTMs, GRUs, and convolutional sequence models. Although these systems have achieved impressive empirical performance, none exhibit the combination of function-preserving scaling, attractor stability, staged consolidation, and lineage formation demonstrated by GLPG.

2.1 Scaling in Conventional Neural Architectures

Early neural language models (MLPs, RNNs, LSTMs) do not support architectural scaling without full retraining. Increasing hidden-layer capacity changes the model’s inductive biases, alters its internal attractor landscape, and typically requires training from scratch. Downscaling is even more destructive: reducing capacity collapses attractors, erases learned structure, and produces models that bear little resemblance to their larger predecessors. No lineage or inheritance is possible.

Transformers introduced more predictable scaling laws, but only in the statistical sense. Kaplan et al. (2020) and subsequent work showed that transformer performance follows power-law relationships with respect to dataset size, model size, and compute. However, these laws do not imply function-preserving scaling. A larger transformer is not a continuation of a smaller one: it must be trained from scratch, and its internal representations differ substantially. Attempts at function-preserving scaling—such as Net2Net transformations, layer expansion, or width doubling—have limited applicability and do not preserve semantic manifolds in generative models.

Moreover, transformers exhibit several instability modes under scaling: attention collapse, gradient interference, catastrophic forgetting, embedding drift, and sensitivity to initialization and optimizer dynamics. Small changes in hyperparameters or dataset composition can produce large, unpredictable shifts in output quality. These behaviours contrast sharply with GLPG’s smooth manifold deformation under upscaling, downscaling, and staged growth.

2.2 Distillation, Compression, and Model Surgery

Knowledge distillation (Hinton et al., 2015) compresses large models into smaller ones, but the resulting student model is not a structural continuation of the teacher. Distillation is lossy, requires a separate training process, and does not preserve the teacher’s attractor geometry. Similarly, pruning, quantization, and low-rank factorization reduce model size but degrade semantic structure and do not support reversible scaling.

Net2Net transformations (Chen et al., 2016) introduced function-preserving network expansion and contraction for feedforward models, but these techniques do not apply to generative architectures with recurrent or attention-based dynamics. Even when applicable, Net2Net transformations preserve only local function approximations, not global attractor manifolds or semantic world models.

GLPG differs fundamentally: downscaling acts as self-distillation, compressing the manifold without erasing canonical attractors, while upscaling expands capacity without cognitive reset. This bidirectional, reversible scaling has no analogue in existing language model architectures.

2.3 Staged Growth and Lineage Formation

Staged or curriculum-based training has been explored in various contexts, including progressive nets, growing neural gas, and curriculum learning. However, these methods do not produce hereditary attractor basins or multi-generational cognitive lineages. Transformers trained with curricula do not inherit structured attractors from smaller ancestors; they simply converge more quickly or more stably.

The closest analogue is the Toy Academy work, which demonstrated staged growth for function learning. GLPG extends this idea from functions to worlds: staged training produces semantic clusters that propagate across generations, enabling genuine lineage formation. No existing language model architecture exhibits this behaviour.

2.4 Transfer Learning and Canon Dynamics

Transfer learning in transformers typically results in catastrophic forgetting or unstable interference between domains. Multi-domain training requires careful balancing, adapter layers, or mixture-of-experts routing to avoid collapse. Even then, semantic interference remains a major challenge.

GLPG’s transfer learning behaviour is qualitatively different. The attractor manifold reorganises in a deterministic, canon-dominant manner: vocabulary persists, but semantic activation follows the dataset with the strongest attractor structure. This is not catastrophic forgetting but a stable, predictable cognitive overwrite analogous to biological learning dynamics.

2.5 Summary

Across all prior architectures, scaling is either destructive (downscaling), non-preserving (upscaling), or requires full retraining (transformers). No existing system supports reversible, function-preserving scaling; no system maintains stable attractor geometry across architectural transformations; no system exhibits hereditary semantic clusters; and no system demonstrates canon-dominant transfer learning.

GLPG therefore occupies a unique position in the landscape of neural architectures: a minimal, interpretable, evolution-capable cognitive substrate whose scaling behaviour has no precedent in conventional language models.

3 Effects of Grammar Fixing

The original GLPG engine included an optional grammar-fixing layer designed to correct surface-level linguistic mistakes after generation. Although well-intentioned, this mechanism introduced a critical flaw: it operated outside the canonical grammar-lemma graph and was therefore capable of producing tokens that did not exist in the canonical domain. As a result, grammar fixing could break canon, distort lemma embeddings, and introduce semantic drift that did not originate from the GLPG model itself.

Observed Grammar Fix Behaviour

During early experiments, the grammar fixer produced several classes of corrections:

1. **Verb normalization:** *woulds* → *would*, *coulds* → *could*, *musts* → *must*, *oughts* → *ought*.
2. **Canon-breaking substitutions:** These were the most problematic, replacing canonical tokens with unrelated words: *Zaidie* → *aide*, *murgatroyd* → *purgatory*, *dejeuner* → *Lejeune*, *Weena* → *Deena*, *morlocks* → *warlocks*, *saltpetre* → *saltwater*, *astronef* → *estrone*. Such substitutions violate the canon-closed nature of GLPG and corrupt the learned lemma geometry.
3. **Contraction repair:** *did n’t* → *didn’t*, *husband ’s* → *husband’s*, and similar apostrophe handling.
4. **Capitalization changes:** *martian* → *Martian*, *mars* → *Mars*, *jupiter* → *Jupiter*, *andrew* → *Andrew*, *mrs. sends* → *Mrs. Sends*.
5. **Singular/plural and conjugation adjustments.**
6. **Minor lexical corrections:** *too* → *to*, *a* → *an*, *an* → *a*, *its* → *it’s*.
7. **Punctuation changes:** Adding commas or replacing periods with question marks.
8. **English variant normalization:** *grey* → *gray*, *centre* → *center*, *marvellous* → *marvelous*, *colour* → *color*, *recognise* → *recognize*.
9. **Hyphenation and word-joining:** *south eastern* → *south-eastern*, *any more* → *anymore*.
10. **Insertion of missing verbs or pronouns:** e.g. adding *is* or *I*.

11. Word removal: e.g. *these their* → *these*.

Although many of these corrections were linguistically reasonable, they were structurally unsafe: the grammar fixer could introduce tokens not present in the canonical dataset, thereby breaking GLPG’s core guarantee of canon-closed generation.

Deterministic Fixing Rules

To eliminate canon-breaking behaviour, the grammar fixer has been replaced by a small set of deterministic, safe post-processing rules. These rules correct only trivial surface-level issues and never introduce new tokens. The current fixing layer applies:

- contraction normalization: " n't " → " not ", " ' " → " ' ",
- simple lexical repairs: " woulds " → " would ", " a a " → " an a ",
- specific phrase corrections: "I going" → "I’m going",
- capitalization of property names (e.g. " america " → " America ").

These rules are intentionally minimal and deterministic. They preserve the canonical token inventory, avoid semantic drift, and ensure that the model output is evaluated exactly as generated, without an additional stochastic or corpus-driven correction layer.

Removal of Grammar-Based Postprocessing

The optional grammar checker remains available for diagnostic purposes, but it is no longer used during generation or evaluation. Any discrepancy between the deterministic fixes and the grammar checker output is logged for inspection. This logging mechanism allows detection of potential structural issues without modifying the model output.

Summary

The removal of the original grammar fixer restores GLPG’s core architectural principle: **all generated text must remain canon-closed**. The new deterministic fixing layer ensures surface-level consistency while preserving the integrity of the learned lemma geometry, the paging structure, and the attractor manifold.

4 Output Scoring and Evaluation

To compare cognitive behaviour across scaling experiments, every model generates a fixed set of outputs: twenty independent stories of twenty lines each, produced in both generative modes supported by GLPG (paged graph traversal and pure MLP mode). These stories are evaluated by an external LLM, providing an objective and reproducible numerical metric for model quality.

4.1 External LLM Evaluator

All evaluations are performed using the Mistral 7B v0.3 instruct model, operating as a blind, architecture-agnostic assessor. The evaluator receives only the generated story text and applies a calibrated scoring regime from 1 to 100. The prompt is:

“Give a score between 1 and 100 for the following short story. The story is written on a moderate level. A score below 50 is perfectly acceptable. Assume the average moderate story scores around 50. Only exceptional stories should score above 80. Evaluate using the following criteria: content, coherence, readability, originality, creativity, style, structure. Do not explain the score.”

This calibrated regime ensures that mid-quality stories cluster around 50, weak stories fall below 40, and only highly coherent or stylistically rich stories exceed 80. The evaluator is invoked twice per story to reduce stochastic variance.

4.2 Evaluation Pipeline

The evaluation process follows three steps:

1. **Story extraction.** Each output file is segmented into stories of at least twenty lines. Shorter fragments are discarded. A maximum of one thousand stories may be processed, though all experiments in this paper use exactly twenty.
2. **Dual scoring.** For each story, the evaluator produces two independent scores:

$$s_1, s_2 \in [1, 100].$$

If either score falls outside the valid range (due to malformed output), the evaluator retries until a valid score is obtained.

3. **Score consolidation.** From the two scores, three per-story values are computed:

$$s_{\min} = \min(s_1, s_2), \quad s_{\max} = \max(s_1, s_2), \quad s_{\text{avg}} = \left\lfloor \frac{s_1 + s_2 + 1}{2} \right\rfloor.$$

These values capture evaluator variance and provide a representative score for the story.

4.3 Score Aggregation

To obtain a compact and comparable metric across all scaling experiments, the twenty stories generated by each model are aggregated into six values:

- **Total Min:** $\sum s_{\min}$ — global lower-bound quality.
- **Total Max:** $\sum s_{\max}$ — global upper-bound quality.
- **Total Avg:** $\sum s_{\text{avg}}$ — representative global quality.
- **Min:** $\min s_{\text{avg}}$ — weakest attractor basin.
- **Max:** $\max s_{\text{avg}}$ — strongest attractor basin.
- **Median:** median of all s_{avg} — typical attractor strength.

These six values form a stable cognitive fingerprint of each model. The total scores measure global attractor strength and semantic coherence; the per-story min, max, and median measure manifold spread, attractor depth, and narrative stability. Together, they allow direct comparison between models of different sizes, training depths, and generational lineages.

4.4 Rationale

GLPG is a deterministic-structural yet semantically emergent architecture. Its behaviour is best captured not by classical statistical metrics, but by measures that reflect attractor geometry, semantic basin depth, and manifold spread. The six aggregated values introduced here provide a compact and interpretable representation of these cognitive properties, enabling quantitative analysis of scaling behaviour across all experiments in this paper.

4.5 Comparison with Conventional Language Model Evaluation

Conventional language models are typically evaluated using statistical benchmarks such as perplexity, next-token prediction accuracy, masked-token recovery, or large-scale QA and reasoning datasets. These metrics assume a probabilistic generative mechanism in which the model attempts to approximate the statistical distribution of a corpus. Such evaluations are appropriate for transformer-based architectures, where semantic behaviour emerges from high-dimensional attention patterns and large-scale correlation.

GLPG does not operate in this regime. Its generative behaviour is governed by a deterministic grammar–lemma graph, explicit paging geometry, and a compact monolithic MLP that injects semantic content into a canon-closed structural substrate. As a result, GLPG is not a statistical language model: it does not attempt to predict corpus-level token frequencies, nor does it optimize likelihood over a distribution. Instead, GLPG produces structured prose constrained by a fixed canonical domain. The quality of its output can therefore never exceed the quality of the canonical dataset itself; the dataset defines an upper bound on semantic richness, stylistic variation, and narrative complexity.

Because GLPG is canon-driven rather than distribution-driven, conventional LM benchmarks provide little insight into its cognitive behaviour. Perplexity cannot be meaningfully computed on a model whose transitions are filtered by deterministic grammar rules and paging constraints. Masked-token recovery is incompatible with GLPG’s explicit routing and lemma geometry. Reasoning and QA benchmarks assume open-domain semantic generalization, whereas GLPG operates within a closed canonical world.

For these reasons, the present paper adopts an external LLM-based evaluation method rather than statistical LM metrics. The numerical scoring regime introduced here captures the aspects of GLPG that matter: attractor stability, manifold coherence, narrative flow, and stylistic consistency. These properties reflect GLPG’s architectural identity as a deterministic cognitive substrate rather than a probabilistic language model. The evaluation method therefore aligns with GLPG’s design principles and provides a meaningful basis for comparing scaling behaviour across models and generations.

4.6 Comparison with Human Evaluation

Human linguistic ability is traditionally assessed not through statistical benchmarks, but through qualitative scoring of written output. In educational and linguistic settings, a short essay or composition is evaluated by another human using criteria such as content, coherence, readability, creativity, style, and structural clarity. These criteria closely mirror the dimensions used in the LLM-based evaluator adopted in this paper. Human language competence is therefore measured through holistic judgement rather than through token-level prediction accuracy or probabilistic metrics.

Conventional language-model benchmarks do not align with this human-centric evaluation framework. Perplexity assumes that a model should approximate corpus-level token frequencies; QA benchmarks assume broad factual knowledge and open-domain reasoning; masked-token recovery assumes statistical reconstruction of missing information. None of these assumptions apply to human language production, where quality is judged by narrative flow, stylistic consistency, and conceptual coherence rather than by statistical likelihood.

GLPG operates in a similar regime. As a canon-driven architecture, GLPG generates text within a closed structural and semantic domain defined by its canonical dataset. The quality of its output can therefore never exceed the quality of the canon itself; the dataset imposes an upper bound on semantic richness and stylistic variation. This mirrors human writing: a person cannot produce knowledge or stylistic nuance beyond what they have been exposed to. GLPG’s generative behaviour is thus constrained by its learned world model rather than by open-domain statistical generalization.

For these reasons, human-style evaluation is more appropriate for GLPG than conventional LM benchmarks. The external LLM assessor used in this paper functions analogously to a human grader: it reads the generated story, applies qualitative criteria, and assigns a holistic score. This approach captures the aspects of GLPG that matter—attractor stability, narrative coherence, semantic consistency, and stylistic fidelity—while avoiding metrics that presuppose statistical language modelling.

In the broader context of AGI research, such human-aligned evaluation may become increasingly

important. As cognitive substrates evolve beyond probabilistic token predictors, naturalistic assessment of narrative quality, conceptual structure, and stylistic competence provides a more meaningful measure of emergent intelligence. GLPG’s evaluation method therefore represents a step toward future AGI benchmarks that reflect human linguistic judgement rather than purely statistical performance.

5 What is Scaling

GLPG supports architectural scaling: the hidden-layer capacity of the monolithic MLP can be increased or decreased while preserving the learned cognitive manifold. This behaviour is unusual among language models. Conventional architectures require full retraining when capacity changes, because their internal representations are statistically entangled. GLPG, however, separates structure (the deterministic grammar–lemma graph and paging geometry) from semantics (the learned lemma embeddings and MLP projection). As a result, the canonical world model can be preserved across architectural transformations.

Scaling therefore refers to the process of transforming a trained GLPG model into a larger or smaller version while retaining its attractor basins, canonical structure, and semantic geometry.

5.1 Conceptual Overview

GLPG scaling is governed by two principles:

1. **Structural invariance.** The grammar–lemma graph, paging geometry, and learned lemma embeddings remain unchanged during scaling. These components form the cognitive DNA of the model.
2. **Semantic projection.** The monolithic MLP is replaced by a larger or smaller network whose weights are derived from the original model through conservative embedding (upscaling) or projection (downscaling). This preserves the semantic manifold while adjusting its expressive capacity.

Because the canonical world model is preserved, scaling does not reset the model. Instead, the model continues training from a transformed but structurally identical cognitive substrate.

5.2 Upscaling

Upscaling increases the hidden-layer size of the monolithic MLP. The goal is to expand the semantic manifold without erasing existing attractors. GLPG accomplishes this by embedding the old weight matrices into the top-left corner of larger matrices and initializing all new parameters with extremely small values. This ensures that the expanded model behaves like the original model at the start of training, while providing additional capacity for semantic growth.

Formally, given an old hidden size H and a new size $H' > H$, each linear layer is widened:

$$W' = \begin{pmatrix} W & \varepsilon \\ \varepsilon & \varepsilon \end{pmatrix}, \quad b' = (b, \varepsilon),$$

where ε is tiny noise (e.g. 10^{-5}). The old cognitive manifold is preserved in the upper-left block, while the new dimensions begin as neutral extensions of the existing space.

This mechanism allows a small, heavily trained model to seed a larger model that requires only light additional training to achieve high performance. Upscaling therefore acts as capacity-expansion without cognitive reset.

5.3 Downscaling

Downscaling reduces the hidden-layer size. The goal is to compress the semantic manifold into a smaller core while preserving its canonical structure. GLPG accomplishes this by projecting each large weight matrix onto its top-left block:

$$W' = W[:, H', : H'], \quad b' = b[:, H'].$$

This operation extracts the most stable portion of the attractor manifold. Because GLPG’s semantic geometry is strongly shaped by paging and deterministic routing, the first H' dimensions typically contain the core attractor basins. Downscaling therefore acts as cognitive compression.

Downscaled models often exhibit higher local coherence and stronger canonical pressure, even though their expressive capacity is reduced. This behaviour mirrors biological systems: smaller brains can retain stable behavioural patterns while losing higher-level variation.

5.4 Algorithmic Description

The upscaling and downscaling procedures are implemented directly in the GLPG reference engine. The following excerpt illustrates the core mechanism:

Upscaling: embed old weights into larger matrices; initialize new parameters with tiny noise.

Downscaling: project large matrices onto smaller blocks; retain the first H' dimensions.

These operations preserve the functional behaviour of the model while adjusting its capacity. Training can continue immediately after scaling, allowing the model to refine its expanded or compressed manifold without losing previously learned structure.

5.5 Implications

Scaling reveals that GLPG behaves as a cognitive substrate rather than a statistical model. Its semantic manifold is stable, hereditary, and structurally grounded. Upscaling inflates the manifold; downscaling compresses it; staged training produces generational inheritance. The baseline model therefore serves not only as a reference point, but as the progenitor of an entire lineage of scaled models explored in the following sections.

6 What is Staged Training

Staged training is a multi-generational growth procedure in which a GLPG model is trained across several architectural capacities in sequence. Instead of training a single large model from scratch, the model is grown through a lineage of progressively larger hidden-layer sizes. Each stage inherits the canonical world model, lemma embeddings, and paging geometry from the previous stage, ensuring continuity of the attractor manifold.

This approach mirrors biological and cognitive development: a small system first consolidates its core attractors, after which larger systems expand and refine the inherited semantic manifold. Staged training therefore provides a controlled mechanism for cognitive growth.

6.1 Conceptual Overview

The example staged training pipeline consists of four phases:

1. **Stage 1: Smaller model (384 units).** The model begins with low capacity, forcing strong attractor consolidation. This stage forms the “genetic core” of the lineage.
2. **Stage 2: Baseline-scale model (768 units).** The model expands its semantic manifold while retaining the Stage 1 attractors. New attractor pockets emerge.
3. **Stage 3: Larger model (1152 units).** The manifold becomes richer and more expressive. This stage often produces the most balanced cognitive behaviour.

4. **Stage 4: Largest model (1536 units).** The model reaches full capacity. If trained sufficiently, this stage yields a mature attractor manifold.

Each stage begins from the scaled version of the previous stage rather than from random initialization. This ensures hereditary transfer of attractors and prevents cognitive reset.

6.2 Epoch Allocation

Total training epochs are distributed across stages using fixed multipliers:

$$\text{Multipliers} = [4, 3, 2, 1].$$

Given a total epoch budget E , each stage receives:

$$E_i = \left\lfloor \frac{E}{4 + 3 + 2 + 1} \right\rfloor \times \text{Multiplier}_i.$$

For example, with $E = 200,000$ epochs as budget:

$$E_1 = 80,000,$$

$$E_2 = 60,000,$$

$$E_3 = 40,000,$$

$$E_4 = 20,000,$$

yielding a total of 200,000 epochs.

This allocation reflects the cognitive principle that smaller models require more training to stabilize their attractor basins, while larger models require fewer epochs to refine inherited structure.

6.3 Algorithmic Description

Staged training is implemented in the GLPG engine through two operations:

1. **Scaling.** The model is upscaled or downscaled to the next hidden size using the function-preserving transformations described in Section 5. The canonical world model remains unchanged.
2. **Training.** The scaled model is trained for E_i randomized epochs using the same curriculum as the baseline model. This refines the inherited attractor manifold at the new capacity.

The following pseudocode summarizes the procedure:

```
for each stage  $i$ :   scale model to hidden size  $H_i$    train model for  $E_i$  epochs   save model as
next-stage ancestor
```

This process yields a lineage of models, each inheriting and expanding the attractor geometry of its predecessor.

6.4 Rationale

Staged training exploits a key property of GLPG: its attractor manifold is stable under architectural scaling. A small model can be trained until its attractors are deeply consolidated. Upscaling then embeds this manifold into a larger semantic space, allowing new attractors to form without erasing the old ones. The result is a cognitively richer model that requires far fewer epochs than a large model trained from scratch.

This behaviour is analogous to biological development: early consolidation, followed by capacity expansion, followed by refinement. Staged training therefore provides a computational analogue of cognitive growth and inheritance within the GLPG architecture.

7 Experimental Setup

This section defines the canonical datasets, hyperparameters, and baseline model configuration used throughout all scaling experiments. The baseline model serves as the reference point against which upscaling, downscaling, staged training, and cross-dataset inheritance are evaluated.

7.1 Datasets

All experiments use canonical grammar–lemma corpora derived from two complete Gutenberg novels. These datasets provide compact yet structurally rich narrative biomes suitable for observing GLPG’s attractor dynamics. Both datasets exhibit regular syntactic cadence and constrained vocabulary, making them ideal substrates for studying GLPG’s semantic manifold formation and scaling behaviour.

7.1.1 A Honeymoon in Space

- 357 KB raw text
- 709 KB canonical grammar–lemma tokens
- 54,039 unique token-transition samples
- 685 sequences

7.1.2 The Time Machine

- 180 KB raw text
- 458 KB canonical grammar–lemma tokens
- 35,642 unique token-transition samples
- 238 sequences

7.2 Hyperparameters

All models share the same structural hyperparameters unless otherwise specified:

- **Input vector (124 dimensions):** 44 CSL features (current token, sequence index, line index), 48 token-history features, 32 recurrent narrative-memory features
- **Output embedding:** 128-dimensional learned lemma embedding
- **Baseline model hidden size:** 768 units
- **Paging geometry:** 1024 pages with a maximum of 8 lemma inputs per page (restricting the candidate token space and improving routing determinism)

These hyperparameters define the canonical GLPG manifold width used throughout the scaling experiments.

7.3 Baseline Models

7.3.1 A Honeymoon in Space

- Model size: 39 MB
- Training epochs: 100,000
- Learned transitions: 7,881,963

7.3.2 The Time Machine

- Model size: 38 MB
- Training epochs: 50,000
- Learned transitions: 7,539,465

7.3.3 Role of the Baseline Models

The baseline model defines the canonical attractor manifold for the domain. Scaling experiments—upscaling, downscaling, cross-dataset inheritance, and staged training—are evaluated relative to this baseline. Its attractor geometry, semantic richness, and narrative stability serve as the cognitive reference against which lineage growth and capacity-driven deformation are measured.

8 Scaling Examples

This section illustrates the behaviour of GLPG under direct architectural scaling. All experiments begin from a fully trained baseline model and apply either upscaling or downscaling transformations as defined in Section 5. The goal of this section is purely descriptive: to document the scaling procedures, training budgets, and resulting model geometries. Quality evaluation of these scaled models is presented separately in Section 11.

8.1 Upscaling: *A Honeymoon in Space*

Upscaling expands the hidden-layer capacity of the monolithic MLP while preserving the canonical grammar–lemma graph, paging geometry, and learned lemma embeddings. Three upscaling scenarios were tested on the *Honeymoon in Space* corpus.

(1) Baseline (768) → Large (3072) A fourfold expansion of hidden-layer width.

- Additional training: 50,000 epochs
- Learned transitions: 11,838,669
- Training time: 20,988 s
- Model size: 450 MB

(2) Stage 4 (1536) → Large (3072) A twofold expansion starting from the Stage 4 model of the staged lineage.

- Additional training: 10,000 epochs
- Learned transitions: 16,548,259
- Training time: 4,593 s
- Model size: 450 MB

(3) Large (3072) from scratch A reference model trained directly at full capacity.

- Training: 10,000 epochs
- Learned transitions: 790,135
- Training time: 5,115 s
- Model size: 450 MB

These three configurations provide a controlled comparison between capacity expansion from consolidated models and direct training at full scale.

8.2 Downscaling: *A Honeymoon in Space*

Downscaling compresses the semantic manifold into a smaller hidden-layer width while preserving canonical structure. All downscaling experiments use 5,000 epochs of additional training.

(1) Baseline (768) → Small (384) A halving of hidden-layer width.

- Additional training: 5,000 epochs
- Learned transitions: 8,282,931
- Training time: 198 s
- Model size: 17 MB

(2) Stage 4 (1536) → Small (384) A fourfold compression starting from the Stage 4 model.

- Additional training: 5,000 epochs
- Learned transitions: 16,153,354
- Training time: 165 s
- Model size: 17 MB

(3) Small (384) from scratch Reference model trained directly at small capacity.

- Training: 5,000 epochs
- Learned transitions: 393,775
- Training time: 170 s
- Model size: 17 MB

These configurations allow comparison between compressed manifolds and models trained natively at reduced capacity with a low amount of epochs.

8.3 Downscaling: *The Time Machine*

A second dataset was used to verify that downscaling behaviour generalizes across domains. All experiments use 3,000 epochs of additional training.

(1) Baseline (768) → Small (384)

- Additional training: 3,000 epochs
- Learned transitions: 7,994,909
- Training time: 169 s
- Model size: 16 MB

(2) Stage 4 (1536) → Small (384)

- Additional training: 3,000 epochs
- Learned transitions: 15,418,648
- Training time: 166 s
- Model size: 16 MB

(3) Small (384) from scratch

- Training: 3,000 epochs
- Learned transitions: 451,269
- Training time: 181 s
- Model size: 16 MB

Summary

The scaling experiments documented in this section establish the structural behaviour of GLPG under both expansion and compression of hidden-layer capacity. Upscaling preserves the inherited semantic manifold while enabling further refinement; downscaling compresses the manifold while retaining core attractors. The quality implications of these transformations are analysed in Section 11.

9 Staged Learning Examples

Staged learning is a multi-generational training procedure in which a GLPG model is grown through a sequence of progressively larger hidden-layer capacities. Each stage inherits the canonical grammar-lemma graph, paging geometry, and learned lemma embeddings from the previous stage, ensuring continuity of the attractor manifold. This section documents the staged training runs performed on two canonical datasets. Quality evaluation of these staged models is presented separately in Section 11.

9.1 A Honeymoon in Space

The staged lineage for *A Honeymoon in Space* uses a total budget of 200,000 epochs. Training proceeds through four stages: 384 → 768 → 1152 → 1536 hidden units. The total training time is 8,445 s.

For comparison, training a single 1536-unit model for the full 200,000 epochs would require:

$$200 \times 1726 \text{ s} / 20 = 17,260 \text{ s},$$

which is more than twice as slow as the staged procedure. Staged learning therefore provides a substantial computational advantage while preserving and expanding the inherited attractor manifold.

Hidden Size	Epochs	Transitions	Time (s)	Model Size (MB)
384	80,000	6,296,181	2,245	17
768	60,000	11,031,624	2,324	39
1152	40,000	14,176,395	2,149	73
1536	20,000	15,756,454	1,726	121

Table 1: Staged learning lineage for *A Honeymoon in Space*.

9.2 The Time Machine

The staged lineage for *The Time Machine* uses a total budget of 100,000 epochs. Training proceeds through the same four capacities: $384 \rightarrow 768 \rightarrow 1152 \rightarrow 1536$ hidden units. The total training time is 8,139 s.

Training a single 1536-unit model for the full 100,000 epochs would require:

$$100 \times 1637 \text{ s} / 10 = 16,370 \text{ s},$$

again more than twice the staged training time. This confirms that staged learning provides consistent computational benefits across datasets.

Hidden Size	Epochs	Transitions	Time (s)	Model Size (MB)
384	40,000	5,967,629	2,321	16
768	30,000	10,464,670	2,142	38
1152	20,000	13,456,118	2,040	73
1536	10,000	14,974,968	1,637	121

Table 2: Staged learning lineage for *The Time Machine*.

Summary

Across both datasets, staged learning provides:

- significant reductions in total training time,
- stable inheritance of the attractor manifold across stages,
- progressive expansion of semantic capacity,
- and consistent structural behaviour across narrative biomes.

The quality implications of these staged lineages are analysed in Section 11.

10 Transfer Learning Examples

Transfer learning in GLPG provides a direct window into the behaviour of its attractor manifold. Because GLPG is a canon-closed architecture whose semantic geometry is shaped by deterministic paging and learned lemma embeddings, mixing datasets does not simply combine stylistic features. Instead, transfer learning reveals how strongly each dataset imprints its attractor basins onto the manifold, and how these basins reorganise when a second dataset is introduced.

A hybrid model always retains the combined vocabulary of both datasets, but the active canon—the semantic and structural behaviour expressed during generation—depends entirely on the relative strength of the attractor basins formed during training.

10.1 Honeymoon to Time Machine

The first transfer experiment begins from a fully trained *Honeymoon in Space* baseline model (768 hidden units, 100,000 epochs). The model is then trained for an additional 30,000 epochs on *The Time Machine*, combined with an upscaling step to 1536 hidden units.

- Baseline: 768 hidden units, 100,000 epochs
- Additional training: 30,000 Time Machine epochs
- Upscale: 1536 hidden units
- Learned transitions: 12,388,365
- Training time: 5,910 s
- Model size: 126 MB

Despite the mixed vocabulary, the resulting hybrid model expresses almost entirely *Time Machine* canon. Only isolated lemma tokens from *Honeymoon in Space* remain, without any of their original semantic or structural context. The attractor manifold has been fully reshaped toward the second dataset.

10.2 Time Machine to Honeymoon

The second transfer experiment begins from a fully trained *Time Machine* baseline model (768 hidden units, 50,000 epochs). The model is then trained for an additional 60,000 epochs on *A Honeymoon in Space*, combined with an upscaling step to 1536 hidden units.

- Baseline: 768 hidden units, 50,000 epochs
- Additional training: 60,000 Honeymoon epochs
- Upscale: 1536 hidden units
- Learned transitions: 12,265,834
- Training time: 5,708 s
- Model size: 126 MB

In this direction, the hybrid model expresses almost entirely *Honeymoon in Space* canon. Only a few lemma tokens from *The Time Machine* remain, again without semantic activation. The attractor manifold is reshaped toward the second dataset just as strongly as in the previous experiment.

10.3 Transfer Learning Canon Dynamics

These experiments reveal a fundamental cognitive property of GLPG: **transfer learning is canon-dominant**. The dataset with the strongest attractor structure determines the active canon of the hybrid model, regardless of the combined vocabulary.

Several dynamics emerge:

- **Asymmetric inheritance.** The semantic manifold reorganises toward the second dataset. Training exclusively on domain B after domain A causes A to be largely overwritten by B.
- **Attractor dominance.** The dataset with stronger syntactic cadence, more stable lemma clusters, and higher canon pressure dominates the manifold. In practice, *A Honeymoon in Space* exhibits stronger attractors than *The Time Machine*, but in both directions the second dataset fully determines the resulting canon.

- **Lemma persistence without semantics.** Vocabulary from the first dataset remains present in embedding space, but without semantic activation. These tokens appear only as isolated artefacts, not as coherent narrative structures.
- **Cognitive overwriting.** This behaviour mirrors biological cognitive systems: training a human or animal exclusively in domain B causes domain A to fade. Previously consolidated attractors weaken when new attractors are repeatedly reinforced.
- **Consolidation requires repetition.** To maintain competence in both domains A and B, both must be trained repeatedly. Without continued exposure, one domain will dominate and overwrite the other.

10.4 Comparison with Conventional Language Models

Transformers exhibit a superficially similar phenomenon known as *catastrophic forgetting*, driven by:

- gradient overwriting,
- loss-driven collapse,
- embedding drift,
- attention-pattern overwriting.

However, the mechanism in GLPG is fundamentally different. Transformers forget because new gradients overwrite statistical correlations. GLPG forgets because new attractor basins reorganise the semantic manifold in a deterministic, interpretable way.

Where transformers lose information unpredictably, GLPG exhibits:

- predictable attractor dominance,
- traceable manifold reshaping,
- canon-closed overwriting,
- cognitively interpretable inheritance dynamics.

This makes GLPG’s transfer-learning behaviour not a failure mode, but a hallmark of its identity as a minimal cognitive substrate.

Summary

Transfer learning in GLPG does not blend canons. It reveals how semantic attractors compete, reorganise, and dominate within a unified manifold. The second dataset consistently overwrites the first, unless both are trained repeatedly. This behaviour aligns with biological cognition and distinguishes GLPG from conventional language models, reinforcing its role as a structurally grounded and cognitively interpretable substrate.

11 Trained Models Quality Evaluation

11.1 A Honeymoon in Space

The *Honeymoon in Space* evaluation comprises fourteen trained models, each assessed in both generative modes supported by GLPG: paged graph traversal (P) and pure MLP mode (M). This yields twenty-eight independent outputs, each scored by the external LLM evaluator using the calibrated regime described in Section 4. The resulting metrics provide a detailed view of how scaling, staged learning, and transfer

learning affect the stability and expressiveness of the Honeymoon attractor manifold. The output size provides an indication of the average sentence length.

Model	Mode	KB	Total Min	Total Max	Total Avg	Min	Max	Median
base	P	23	694	878	793	21	51	40
xl new	P	20	609	727	672	25	42	33
xl base	P	24	686	876	785	31	52	39
xl stage4	P	26	754	918	839	29	53	41
small new	P	18	650	831	746	28	48	36
small base	P	24	796	957	882	32	58	45
small stage4	P	21	733	944	844	30	60	43
stage1	P	23	708	869	792	26	55	39
stage2	P	22	697	909	806	28	48	42
stage3	P	24	699	928	820	29	63	40
stage4	P	25	664	884	778	31	48	40
hybrid	P	22	681	839	767	30	51	38
base	M	23	717	907	817	30	54	39
xl new	M	18	572	790	684	27	46	33
xl base	M	25	658	805	738	23	57	37
xl stage4	M	28	685	923	810	27	57	39
small new	M	16	611	727	672	20	45	35
small base	M	25	694	927	815	22	58	41
small stage4	M	21	669	846	763	28	52	38
stage1	M	23	627	744	689	20	43	35
stage2	M	22	693	835	770	26	49	39
stage3	M	26	651	870	766	25	49	38
stage4	M	25	680	846	768	26	57	38
hybrid	M	22	700	858	785	25	52	41

Table 3: Quality evaluation of all Honeymoon models across paged (P) and pure MLP (M) modes.

Ranking and Interpretation To obtain a single comparative score per model, the paged and MLP totals are summed. The resulting ranking reveals several notable cognitive and architectural patterns:

1. **small base: 1697** — The strongest model overall. A downscaled version of the baseline exhibits exceptionally stable attractors and highly coherent prose. Compression strengthens the manifold.
2. **xl stage4: 1649** — Large model grown through staged lineage. Capacity expansion preserves and amplifies inherited attractors.
3. **base: 1610** — The baseline model remains robust and balanced.
4. **small stage4: 1607** — Downscaled Stage 4 retains strong attractor geometry.
5. **stage3: 1586** — Mid-stage models (Stage 2/3) consistently outperform Stage 4.
6. **stage2: 1576** — Stage 2 and Stage 3 form the optimal cognitive regime.
7. **hybrid: 1552** — Mixed-canon models exhibit more diffuse attractors.
8. **stage4: 1546** — Stage 4 requires deeper training to fully stabilise.
9. **xl base: 1523** — Large model trained directly at full capacity underperforms lineage-grown variants.

10. **stage1: 1481** — Early-stage models show limited semantic development.
11. **small new: 1418** — Small model trained from scratch is under-consolidated.
12. **xl new: 1356** — Large model trained from scratch performs worst; capacity alone does not induce strong attractors.

Discussion Several cognitive dynamics emerge clearly from these results:

- **Compression strengthens attractors.** Downscaled models (e.g., *small base*) often outperform larger models, indicating that GLPG’s semantic manifold becomes more coherent when projected into a smaller space.
- **Staged growth outperforms direct training.** Large models trained from scratch (*xl new*) perform significantly worse than large models grown through lineage (*xl stage4*). This confirms that cognitive consolidation at smaller capacities is a prerequisite for successful expansion.
- **Mid-stage models are optimal.** Stage 2 and Stage 3 consistently produce the strongest balance between semantic richness and structural stability.
- **Hybrid models diffuse the manifold.** Mixed-canon training weakens attractor depth, reducing overall coherence.
- **Paged and MLP modes remain close.** Differences between P and M modes are small, confirming that the monolithic MLP internalises the graph structure.

Overall, the Honeymoon evaluation demonstrates that GLPG’s cognitive behaviour depends strongly on architectural scale, training depth, and lineage. Compression, staged growth, and mid-capacity refinement consistently yield the most stable and expressive attractor manifolds.

11.2 The Time Machine

The *Time Machine* evaluation comprises nine trained models, each assessed in both generative modes supported by GLPG: paged graph traversal (P) and pure MLP mode (M). This yields eighteen independent outputs. As in the previous subsection, all stories are scored by the external LLM evaluator, producing a detailed fingerprint of attractor stability and semantic coherence across architectural scales and training regimes.

Model	Mode	KB	Total Min	Total Max	Total Avg	Min	Max	Median
base	P	24	768	1006	888	23	59	47
small new	P	18	788	998	896	28	64	45
small base	P	23	720	885	806	29	54	39
small stage4	P	23	787	1000	900	33	63	44
stage1	P	23	755	970	866	28	55	44
stage2	P	25	717	923	825	29	57	41
stage3	P	25	763	965	869	29	71	43
stage4	P	25	770	927	853	31	49	44
hybrid	P	25	766	1013	895	33	54	44
base	M	24	706	919	818	29	56	40
small new	M	17	689	940	820	29	52	40
small base	M	23	681	886	788	24	58	36
small stage4	M	23	791	1017	911	35	70	44
stage1	M	21	717	1010	870	29	58	42
stage2	M	24	795	1032	918	31	59	47
stage3	M	25	668	860	767	25	52	39
stage4	M	26	768	987	882	28	65	45
hybrid	M	25	730	923	832	27	52	44

Table 4: Quality evaluation of all Time Machine models across paged (P) and pure MLP (M) modes.

Ranking and Interpretation To compare overall performance, the paged and MLP totals are summed for each model. The resulting ranking reveals several distinctive cognitive patterns:

1. **small stage4: 1811** — The strongest model overall. A downscaled Stage 4 model exhibits exceptionally deep attractor basins and highly coherent prose in both modes.
2. **stage2: 1743** — Stage 2 shows very strong MLP-mode behaviour, confirming that mid-stage models often form the richest semantic manifolds.
3. **stage1: 1736** — Surprisingly strong early-stage performance; attractors form quickly on this dataset.
4. **stage4: 1735** — Full-capacity Stage 4 performs well but requires more epochs to fully stabilise.
5. **hybrid: 1727** — Mixed-canon training diffuses attractor depth but remains competitive.
6. **small new: 1716** — Under-trained but still relatively strong due to the dataset’s stable cadence.
7. **base: 1706** — The baseline model performs adequately but is not the strongest in this dataset.
8. **stage3: 1636** — Lower MLP-mode performance reduces the combined score.
9. **small base: 1594** — Downscaled baseline performs weakest; attractor formation is incomplete.

Discussion The Time Machine results reveal several cognitive dynamics distinct from the Honeymoon evaluation:

- **Mid-stage models dominate.** Stage 2 and Stage 1 outperform the baseline and even Stage 4, indicating that the Time Machine manifold stabilises early and benefits from moderate capacity.
- **Downscaled Stage 4 is optimal.** Compression again strengthens attractor coherence, producing the strongest model in both modes.

- **MLP mode is highly expressive.** Several models (Stage 2, Stage 1, small stage4) show exceptionally high MLP-mode scores, confirming that the monolithic MLP internalises the dataset’s semantic structure effectively.
- **Hybrid models remain competitive.** Although mixed-canon training diffuses attractor depth, hybrid models still perform near the top of the ranking.

Overall, the Time Machine evaluation demonstrates that GLPG’s cognitive behaviour depends strongly on training depth and lineage, with mid-stage and compressed models consistently yielding the most stable and expressive attractor manifolds for this dataset.

11.3 Summary

Across both canonical datasets, the evaluation results reveal a coherent and interpretable picture of GLPG’s cognitive behaviour under scaling, staged learning, and transfer learning. Despite substantial variation in model size, training depth, and architectural lineage, the overall quality of generated stories remains remarkably stable. Most models achieve median scores around 40 or higher, with occasional outputs reaching moderate quality (scores near 50) and rare outliers approaching 70. Scores above 80—corresponding to exceptional narrative quality—are not observed, consistent with the structural limits imposed by the canonical datasets themselves.

Several global trends emerge:

- **Paged mode is consistently faster and more stable.** Paged graph traversal is more than four times faster than pure MLP mode and exhibits tighter structural cadence. MLP mode remains expressive but computationally expensive.
- **Sentence length correlates with capacity and training depth.** Larger models and models trained for more epochs tend to produce longer sentences, reflecting broader semantic manifolds and deeper attractor basins.
- **Compression strengthens attractors.** Downscaled models often outperform their larger counterparts. Projecting the manifold into a smaller hidden space increases structural coherence and reduces semantic drift.
- **Staged growth outperforms direct training.** Large models trained from scratch perform significantly worse than large models grown through staged lineage. Cognitive consolidation at smaller capacities is a prerequisite for successful expansion.
- **Mid-stage models form the optimal cognitive regime.** Stage 2 and Stage 3 consistently yield the strongest balance between semantic richness and structural stability across both datasets.
- **Hybrid models diffuse attractor depth.** Mixed-canon training produces competitive but slightly weaker results, reflecting a more diffuse semantic manifold.
- **Dataset complexity influences manifold behaviour.** *The Time Machine* models generate faster and often exhibit stronger early-stage attractors, while *A Honeymoon in Space* models show more pronounced benefits from staged growth and compression.
- **Shorter outputs can score higher.** Models producing shorter, structurally simpler sentences often achieve better scores due to reduced narrative complexity and fewer opportunities for semantic drift.

Overall, the evaluation demonstrates that GLPG maintains stable generative quality across a wide range of capacities, training regimes, and datasets. The architecture’s cognitive properties—compression-driven coherence, staged consolidation, attractor-based semantics, and dual-mode generation—produce a robust and interpretable manifold whose behaviour remains consistent even under substantial architectural variation.

12 Textbook Scaling Behaviour

Across all scaling experiments, GLPG exhibits a remarkably consistent and textbook-like developmental pattern. Upscaling monotonically improves semantic richness and attractor depth; downscaling compresses the manifold while preserving canonical structure; tiny models reach a clear capacity limit; lineage-based XL models outperform all baselines; and XL models trained from scratch collapse due to insufficient consolidation. These behaviours occur uniformly across all model sizes, generational paths, and datasets.

Beyond these monotonic scaling laws, GLPG displays an additional and highly unusual property: **structural and semantic stability under variation**. Across hyperparameter sweeps, scaling directions, staged learning regimes, transfer learning experiments, and dual-mode generation, the model’s attractor manifold remains coherent and predictable. Changes in learning rate, batch size, initialization, training depth, or model capacity do not produce chaotic jumps, collapses, or irregularities. Instead, GLPG consistently deforms its manifold in smooth, interpretable ways.

This stability is not typical for language models. Transformer-based architectures often show non-monotonic scaling behaviour, irregular performance jumps, catastrophic forgetting, embedding drift, and sensitivity to initialization and optimizer dynamics. Small changes in hyperparameters or dataset composition can produce large, unpredictable shifts in output quality. GLPG, by contrast, behaves as a low-dimensional cognitive dynamical system: consolidation at smaller capacities stabilizes the attractor manifold, which can then be expanded through upscaling without resetting the learned world model. Downscaling compresses the manifold without destroying canonical structure, and transfer learning re-organizes attractors in a deterministic, canon-dominant manner.

The textbook nature of GLPG’s scaling behaviour arises from its architectural separation between structure and semantics. The deterministic grammar–lemma graph and paging geometry provide a stable backbone, while the monolithic MLP acts as a semantic projector whose capacity can be increased or decreased without entangling the underlying canonical world. Function-preserving scaling therefore yields predictable cognitive deformation: compression, stabilization, expansion, and refinement.

This combination of monotonic scaling laws and cross-regime stability makes GLPG uniquely suitable for staged growth, lineage-based training, and controlled cognitive evolution. The scaling results in this paper demonstrate that GLPG’s manifold behaves not as a statistical correlation space, but as a structured cognitive substrate whose development follows clear, reproducible, and biologically-plausible patterns.

13 Relevance for AGI

GLPG exhibits a constellation of properties that are rare or entirely absent in conventional language model architectures. Most neural language models—MLPs, RNNs, LSTMs, and transformers—do not support function-preserving scaling, do not preserve their attractor manifold under architectural change, and do not maintain stable behaviour under hyperparameter variation, dataset mixing, or staged training. Increasing or decreasing model capacity in such systems typically requires full retraining, and the resulting model is not a continuation of the smaller model but a new statistical system with different inductive biases, different attractors, and different failure modes.

GLPG behaves differently. Its monolithic MLP supports function-preserving hidden-layer doubling: upscaling expands capacity while preserving the attractor manifold; downscaling compresses the manifold without destroying its structure. As a result, GLPG models can grow and shrink without cognitive reset. This behaviour is reinforced by a second, equally unusual property: **structural and semantic stability**. Across hyperparameter sweeps, scaling experiments, staged learning, transfer learning, and dual-mode generation, GLPG’s manifold deforms smoothly and predictably. No chaotic jumps, collapses, or irregularities appear. This stability is not typical for statistical language models and reflects GLPG’s identity as a low-dimensional cognitive dynamical system.

13.1 Cognitive Manifold Preservation

The experiments in this paper demonstrate that GLPG maintains a stable cognitive manifold under both upscaling and downscaling. A larger model is not a new model; it is a more mature version of the smaller model. The canonical attractors—characters, motifs, stylistic rhythms, and world-objects—remain intact across generations. Even after extreme transformations, the canonical elements persist across all stages.

This stability is the foundation for GLPG’s lineage behaviour: models inherit structure, style, and attractors from their ancestors, forming a coherent genealogical line.

13.2 Self-Distillation Through Downscaling

Downscaling acts as a form of self-distillation. Instead of training a new student model on a new dataset, GLPG compresses its own manifold into a smaller semantic core. This core retains the essential attractors of the parent model. Subsequent upscaling expands this core into a larger semantic space, producing a model that is structurally similar to the parent but often more stable and more coherent.

upscaling = capacity expansion, downscaling = self-distillation.

13.3 Lineage and Generational Growth

The scaling experiments show that GLPG supports genuine lineage formation. Models can be trained across multiple generations, each inheriting the attractor manifold of the previous generation. This produces predictable cognitive phases:

- **Undertraining** leads to drift but not collapse.
- **Overtraining** produces hypercoherent baselines.
- **Downscaling** compresses the manifold.
- **Upscaling** inflates the manifold.
- **Generations** can be evaluated and compared.
- **Evolution** becomes visible in the output.

This constitutes an artificial evolutionary line: a computational analogue of developmental growth.

13.4 Transfer Learning and Canon Dominance

Transfer learning reveals another AGI-relevant property: **canon dominance**. When a model trained on dataset A is subsequently trained on dataset B, the attractor manifold reorganises deterministically toward B. Vocabulary from A persists, but its semantic activation disappears. This behaviour mirrors biological cognition: new attractors overwrite old ones unless both domains are repeatedly reinforced.

This is not catastrophic forgetting in the transformer sense. GLPG does not lose information through gradient interference; instead, its attractor basins reorganise in a stable, predictable, canon-closed manner. The dataset with the strongest attractor structure dominates the hybrid manifold. This is a cognitive phenomenon, not a statistical failure mode.

13.5 The XL Scratch Ablation

The XL-from-scratch experiment provides a critical ablation:

1. A 3072-unit model trained from scratch for 10K epochs performs significantly worse than the genealogical XL model.

2. Its attractors differ: fewer canonical motifs, weaker structure, more repetition, and less coherent rhythm.

This demonstrates that lineage is not merely a convenience but a cognitive factor. Capacity alone is insufficient; staged consolidation is required.

13.6 The GLPG Cognitive Consolidation Principle

The experiments support a general principle:

A model’s attractor manifold must be stabilized at lower capacities before architectural upscaling. Upscaling without sufficient consolidation leads to monotonic attractor collapse, even when total parameter count increases. Upscaling with consolidation leads to cognitive growth.

Toy Academy demonstrated that staged growth works for functions. GLPG demonstrates that staged growth works for worlds.

13.7 Towards an AGI Ecology

The behaviour observed in GLPG resembles properties of biological and cultural systems:

- diversity of attractors,
- niche formation,
- cultural evolution,
- redundancy,
- emergent behaviour between agents.

AGI is unlikely to be an individual system. AGI is a collective phenomenon.

You do not build an AGI model. You build an AGI ecology.

GLPG’s scaling, stability, lineage, and self-distillation properties provide a concrete computational mechanism for such an ecology: multiple agents, each with inherited attractors, evolving through staged growth, dataset transfer, compression, expansion, and cross-generational consolidation. GLPG does not behave like a statistical language model; it behaves like a structured cognitive substrate whose development follows clear, reproducible, and biologically plausible patterns.

14 Conclusion

The experiments in this paper demonstrate that GLPG is not merely another language model architecture, but a structurally grounded cognitive substrate whose behaviour is stable, interpretable, and reproducible across a wide range of capacities, datasets, and training regimes. Unlike statistical language models, whose scaling behaviour is often irregular, sensitive, and prone to collapse, GLPG exhibits textbook developmental dynamics: compression strengthens attractors, staged consolidation stabilizes the manifold, upscaling expands semantic capacity without cognitive reset, and transfer learning reorganizes attractors in a deterministic, canon-dominant manner.

Across all evaluations, GLPG maintains a coherent attractor manifold. Small models form stable semantic cores; downscaled models retain essential structure; mid-stage models achieve optimal balance between richness and stability; and lineage-grown XL models outperform scratch-trained variants by a wide margin. Even under extreme architectural transformations, canonical elements persist, stylistic rhythms remain intact, and the manifold deforms smoothly rather than chaotically. This robustness

is unusual for neural language models and reflects the architectural separation between structure and semantics: the grammar–lemma graph provides a stable backbone, while the monolithic MLP acts as a semantic projector whose capacity can be increased or decreased without entangling the underlying world model.

The cognitive implications are significant. GLPG supports function-preserving scaling, self-distillation through downscaling, generational lineage formation, and predictable cognitive growth. Transfer learning reveals canon dominance rather than catastrophic forgetting, and staged training demonstrates that consolidation at lower capacities is a prerequisite for successful expansion. These behaviours mirror properties of biological and cultural systems: inheritance, niche formation, attractor competition, and evolutionary development.

Taken together, the results suggest that GLPG provides a concrete computational mechanism for building not a single AGI model, but an AGI ecology: multiple agents, each with inherited attractors, evolving through staged growth, dataset transfer, compression, expansion, and cross-generational consolidation. GLPG does not behave like a statistical correlation engine. It behaves like a structured cognitive dynamical system whose development follows clear, reproducible, and biologically plausible patterns.

In this sense, GLPG offers a path toward scalable artificial cognition: an architecture capable of growing worlds, preserving minds, and supporting the emergence of complex, multi-agent intelligence.

A Validation of the Numerical Evaluation Method

To verify that the numerical evaluation pipeline produces meaningful and credible results, we performed a cross-check using the best-performing models from both canonical datasets: *A Honeymoon in Space* (small base, paged mode) and *The Time Machine* (small stage 4, paged and MLP modes). For each model, twenty independent stories were generated and scored by an external LLM using the calibrated evaluation prompt described in Section 3. The scoring regime defines moderate-level quality around a median of 50, with scores above 80 reserved for exceptional human-level coherence.

A.1 Honeymoon Cross-Check

The best Honeymoon model (small base, paged mode) produced scores ranging from 32 to 58, with a median of 45 and a total average of 882. The distribution is tightly clustered around the moderate-quality band, with no outliers above 60 and no collapses below 20. This matches the expected behaviour of GLPG’s paged generation: structurally stable, stylistically consistent, but semantically diffuse. The score distribution is monotonic, smooth, and free of anomalies, confirming that the evaluator responds predictably to GLPG’s output characteristics.

A.2 Time Machine Cross-Check

The best Time Machine model (small stage 4) was evaluated in both paged and MLP modes. Paged mode produced scores between 33 and 63 (median 44, total 900), while MLP mode produced scores between 35 and 70 (median 44, total 911). The similarity of the medians demonstrates that the monolithic MLP internalises the graph structure, while the slightly higher maximum scores in MLP mode reflect its tendency to generate longer and locally more coherent sentences. Crucially, both modes remain within the moderate-quality band, with no exceptional scores above 80 and no pathological failures below 20.

A.3 Consistency and Realism

Across both datasets and both generative modes, the evaluation pipeline yields:

- **Realistic score ranges** (30–60), consistent with moderate-level narrative quality.
- **Stable medians** (44–45), reflecting GLPG’s attractor-driven coherence.

- **Plausible maxima** (58–70), indicating occasional strong passages without exceeding the exceptional threshold.
- **Absence of collapses** (no scores below 20), demonstrating GLPG’s structural stability.
- **Dataset-sensitive variation**, with Time Machine producing slightly higher maxima due to its stronger lemma clusters.
- **Mode-sensitive variation**, with MLP mode occasionally producing higher peaks while preserving the same median as paged mode.

These properties confirm that the evaluation method is both *credible* and *diagnostic*. The scoring pipeline does not inflate results, does not compress distributions, and does not produce artefactual extremes. Instead, it yields smooth, interpretable score profiles that align with the known cognitive behaviour of GLPG.

A.4 Conclusion

The cross-check demonstrates that the numerical evaluation pipeline provides a reliable and realistic measure of GLPG’s performance. The scores are consistent with the expected moderate-level quality of GLPG outputs, sensitive to architectural differences, and stable across datasets and generative modes. This validates the use of the external LLM evaluator as an objective metric for assessing cognitive behaviour in GLPG.

B Graph-Based QKV Selection in GLPG

GLPG does not use attention, nor does it rely on the query–key–value mechanism found in transformers. However, the structure of the GLPG language graph naturally induces an analogous triplet. This analogy is not architectural but conceptual: it clarifies which parts of the GLPG pipeline are already deterministic, and which part remains an open challenge for further quality improvements.

B.1 Deterministic K and V in the Language Graph

In GLPG, every lemma is a node in the canonical language graph. Each node has a set of outgoing transitions representing all semantically valid continuations. This yields two deterministic components:

- **K (Key):** the lemma itself, representing *where* we are in the canonical world. The key is not a learned vector but a structural position in the graph.
- **V (Value):** the set of all outgoing transitions from that lemma, representing *what* can happen next. The value is not a dense projection but a deterministic transition set.

Thus, GLPG already provides the “K” and “V” components through its world model. They are not learned; they are part of the canonical structure.

B.2 Q as Context-Derived Motivation

The missing component is the **Q (Query)**, which must determine *why* a particular transition from $V(K)$ should be selected. Unlike transformers, where Q is a learned linear projection of the token embedding, GLPG must derive Q from the input vector and the paging context.

Q must encode:

- attractor pressure,
- narrative intent,

- world-state consistency,
- lemma-embedding similarity,
- and the cognitive phase of the model.

At present, GLPG uses simple context encoders to approximate Q . These encoders work, but they are not yet expressive enough to fully exploit the structure of the graph. Improving Q is therefore one of the main challenges for future GLPG development.

B.3 Why This Analogy Matters

This QKV interpretation is not a transformer claim. It is a structural analogy that highlights the division of labour inside GLPG:

$$K = \text{where}, \quad Q = \text{why}, \quad V = \text{what}.$$

Transformers must learn all three components from scratch. GLPG already has K and V embedded in the canonical graph. Only Q must be inferred from context.

This means that GLPG is inherently closer to a *semantic router* than to an attention mechanism. The model does not search over all tokens; it selects among the transitions that the world model allows. Improving Q will therefore directly improve semantic precision, coherence, and narrative stability.

B.4 Open Challenge

The next major step for GLPG is to strengthen the context encoder so that Q becomes a reliable, high-resolution motivation signal. When Q is sufficiently expressive, GLPG will effectively implement a graph-based QKV selection mechanism:

$$V^* = \operatorname{argmax}_{v \in V(K)} \text{compatibility}(Q, v),$$

where compatibility is determined by attractor matching, canonical consistency, and paging context.

This is not attention. It is a deterministic world model guided by a learned motivational vector. But the structural analogy clarifies what is already solved (K and V) and what remains to be improved (Q).

Improving Q is therefore one of the key challenges for increasing GLPG output quality and for enabling more complex multi-agent and multi-world experiments such as the AGI School.

C Triadic Cosmos Ecosystem Context

The present paper is part of the broader *Triadic Cosmos* ecosystem: a unified research program that develops a compositional ontology of intelligence together with modular, inspectable computational architectures. The ecosystem spans conceptual theory, executable reference implementations, and fully documented empirical demonstrations, forming a coherent framework for studying hybrid intelligent systems.

All materials, extended notes, and evolving documentation are publicly available in the open GitHub project:

<https://github.com/triadic-cosmos>

Within the `triadic-system` repository, a complete reference implementation of the GLPG engine, the narrator pipeline, and all supporting modules is provided under the AGPL-3.0 license. The datasets described in this paper are present within the complementary `triadic-data` repository.

References

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